

1、设多项式 $P_3(x) = 4x^3 + 3x^2 + 2x + 1$, 求其值的秦九韶算法公式为: _____.

解:

$$\begin{aligned} P_3(x) &= 4x^3 + 3x^2 + 2x + 1 \\ &= (4x + 3)x^2 + 2x + 1 \\ &= ((4x + 3)x + 2)x + 1 \end{aligned}$$

2、计算 $f(x) = (\sqrt{2} - 1)^6$, 取 $\sqrt{2} \approx 1.4$, 直接计算 f 和利用下述等式计算, 那一个最好? *

$$\frac{1}{(\sqrt{2} + 1)^6}, (3 - 2\sqrt{2})^3, \frac{1}{(3 + 2\sqrt{2})^3}, 99 - 70\sqrt{2}; *$$

解:

$$(1) f(x) = (\sqrt{2} - 1)^6 = (x - 1)^6$$

$$|f(\sqrt{2}) - f(1.4)| \leq |f'(1.4)| e_a = 6 \times (1.4 - 1)^5 e_a = 0.06144 e_a$$

$$(2) f(x) = \frac{1}{(\sqrt{2} + 1)^6} = (x + 1)^{-6}$$

$$|f(\sqrt{2}) - f(1.4)| \leq |f'(1.4)| e_a = 6 \times (1.4 + 1)^7 e_a = 0.01308 e_a$$

$$(3) f(x) = (3 - 2\sqrt{2})^3 = (3 - 2x)^3$$

$$|f(\sqrt{2}) - f(1.4)| \leq |f'(1.4)| e_a = 3 \times 2 \times (3 - 2 \times 1.4)^2 e_a = 0.24 e_a$$

$$(4) f(x) = \frac{1}{(3 + 2\sqrt{2})^3} = (3 + 2x)^{-3}$$

$$|f(\sqrt{2}) - f(1.4)| \leq |f'(1.4)| e_a = 3 \times 2 \times (3 + 2 \times 1.4)^2 e_a = 0.00531 e_a$$

$$(5) f(x) = 99 - 70\sqrt{2} = 99 - 70x$$

$$|f(\sqrt{2}) - f(1.4)| \leq |f'(1.4)| e_a = 70 e_a$$

由计算结果可知, 使用公式 $\frac{1}{(3 + 2\sqrt{2})^3}$ 计算时误差最小

3. 证明 (1) $\|x\|_1 \leq \|x\|_2 \leq n\|x\|_1$

(2) $\|x\|_1 \leq \|x\|_2 \leq \sqrt{n}\|x\|_1$

证明:

$$(1) \because \|x\|_1 = \sum_{i=1}^n |x_i| \geq \max_{1 \leq i \leq n} |x_i| = \|x\|_\infty$$

$$\text{又 } \|x\|_1 = \sum_{i=1}^n |x_i| \leq n \times \max_{1 \leq i \leq n} |x_i| = n\|x\|_\infty$$

$$\therefore \|x\|_\infty \leq \|x\|_1 \leq n\|x\|_\infty$$

$$(2) \because \|x\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}} \geq \left(\max_{1 \leq i \leq n} |x_i|^2 \right)^{\frac{1}{2}} = \max_{1 \leq i \leq n} |x_i| = \|x\|_\infty$$

$$\text{又 } \|x\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}} \leq \left(n \times \max_{1 \leq i \leq n} |x_i|^2 \right)^{\frac{1}{2}} = \sqrt{n} \max_{1 \leq i \leq n} |x_i| = \sqrt{n}\|x\|_\infty$$

$$\therefore \|x\|_\infty \leq \|x\|_2 \leq \sqrt{n}\|x\|_\infty$$

4. 设 $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, 则 $\|A\|_1 = \underline{\quad}$, $\|A\|_\infty = \underline{\quad}$, $\|A\|_{m_1} = \underline{\quad}$, $\rho(A^T A) = \underline{\quad}$, $\|A\|_2 = \underline{\quad}$.

解:

$$A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| = \max \left\{ \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\} = \frac{\sqrt{2}}{2}$$

$$\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}| = \max \left\{ \sqrt{2}, \frac{1}{\sqrt{2}} \right\} = \sqrt{2}$$

$$\|A\|_{m_1} = \sum_{i=1}^m \sum_{j=1}^n |a_{ij}| = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

$$A^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \quad A^T A = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad |\lambda E - A^T A| = \begin{vmatrix} \lambda - \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & \lambda - \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & \lambda - \frac{1}{2} \end{vmatrix} = \begin{vmatrix} \lambda - 1 & 0 & -\frac{1}{2} \\ 0 & \lambda - \frac{1}{2} & 0 \\ 0 & 0 & \lambda \end{vmatrix}$$

$$\therefore \lambda_1 = 0, \lambda_2 = \frac{1}{2}, \lambda_3 = 1$$

$$\therefore \rho(A^T A) = \max |\lambda_i| = \lambda_3 = 1$$

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \sqrt{\rho(A^T A)} = 1$$

$$\text{故 } \|A\|_1 = \frac{\sqrt{2}}{2}, \|A\|_\infty = \sqrt{2}, \|A\|_{m_1} = \frac{3\sqrt{2}}{2}, \rho(A^T A) = 1, \|A\|_2 = 1$$

5. 设 $\|x\|$ 为 $\mathbb{R}^n$ 空间上的任一向量范数, $P \in \mathbb{R}^{n \times n}$ 是非奇异矩阵,

定义 $\|x\|_P = \|Px\|$,

证明: (1) $\|x\|_P = \|Px\|$ 为 $\mathbb{R}^n$ 空间上的一种向量范数;

(2) $\|A\|_P = \|PAP^{-1}\|$ 为算子范数.

证:

(1) ① 非负性

当 $x=0$ 时, $Px=0$, 则 $\|x\|_P = \|Px\| = 0$

当 $x \neq 0$ 时, P 为非奇异矩阵, $Px \neq 0$, 故 $\|x\|_P = \|Px\| > 0$

$\therefore \|x\|_P = \|Px\| \geq 0$, 当且仅当 $x=0$ 时, $\|x\|_P = 0$

② 齐次性

$$\|\alpha x\|_P = \|P\alpha x\| = \|\alpha Px\| = |\alpha| \|Px\| = |\alpha| \|x\|_P$$

③ 三角不等式

$$\|x+y\|_P = \|P(x+y)\| = \|Px+Py\| \leq \|Px\| + \|Py\| = \|x\|_P + \|y\|_P$$

综上, $\|x\|_P$ 为向量范数

$$(2) \|A\|_P = \max_{x \neq 0} \frac{\|Ax\|_P}{\|x\|_P} = \max_{x \neq 0} \frac{\|PAx\|}{\|Px\|} = \max_{x \neq 0} \frac{\|(PAP^{-1})Px\|}{\|Px\|}$$

$$\text{令 } Px = y, \text{ 则 } \|A\|_P = \frac{\|(PAP^{-1})y\|}{\|y\|} = \|PAP^{-1}\|$$

6. 设 $A \in \mathbb{R}^{n \times n}$, λ 是 A 的任意特征值, 证明 $|\lambda| \leq \sqrt[n]{\|A^n\|}$.

证:

$\because \lambda$ 是 A 的特征值, 那 $Ax = \lambda x$

$$\text{则 } A^n x = \lambda A^{n-1} x = \lambda^2 A^{n-2} x = \dots = \lambda^n x$$

$\therefore \lambda^n$ 是 A^n 的特征值

取向量范数 $\|x\|$ 与所给的矩阵范数相容

$$|\lambda|^n \|x\| = \|\lambda^n x\| = \|A^n x\| \leq \|A^n\| \cdot \|x\|$$

$$\text{即 } |\lambda|^n \leq \|A^n\|. \text{ 即 } |\lambda| \leq \sqrt[n]{\|A^n\|}$$